

DOKUMENTASI TEKNIS APLIKASI GAME "LIFE ON LAND"

(TOP-DOWN TACTICAL ECO-RESTORATION SIMULATOR)

Program Studi Teknik Informatika, Fakultas Ilmu Komputer

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BAB I: PENDAHULUAN DAN ARSITEKTUR SISTEM

1.1 Identitas dan Ringkasan Aplikasi

- **Nama Game:** Life on Land
- **Genre:** Top-Down Cozy Stage-Based Forest Ecosystem Simulator
- **Platform:** PC Standalone (Windows 10/11 x86_64, macOS, Linux) & WebGL
- **Engine & Language:** Unity 6 (6000.0.x) / Unity 2022.3 LTS & C# (.NET Core)
- **Backend & Persistence:** Local JSON (*SaveData.json*) / SQLite & PlayFab Cloud Title Data Services

1.2 Topologi Arsitektur (Two-Tier Hybrid Architecture)

Game *Life on Land* menerapkan arsitektur dua tingkat (Two-Tier Hybrid Architecture) yang memisahkan logika komputasi klien (*Client-Side Monolithic Simulation Engine*) dengan layanan penyimpanan awan (*Cloud Persistence & Global Leaderboard Services*):

1. Client-Side Simulation Layer (Unity C#):

- **Input & Controller Module (`PlayerController.cs`):** Mengelola masukan pergerakan WASD/Arrow, penggunaan stamina, buffer oksigen lokal, serta pemilihan slot Hotbar (1–6).
- **Environmental Matrix Module (`EnvironmentManager.cs` & `GridWorldMatrix.cs`):** Menjalankan simulasi spasial 2D untuk kelembapan tanah (*soil moisture*), tingkat kontaminasi lahan (*corruption state*), dan difusi oksigen atmosfer (O_2) berbasis interval waktu (*tick* 5 detik).
- **Plant Lifecycle FSM Module (`Tree.cs`):** Mengontrol siklus hidup biologis tanaman dari *Seed*, *Sprout*, *Young Tree*, *Mature Tree*, hingga *Withered* menggunakan Finite State Machine (FSM).

2. Backend Services Layer (Microsoft PlayFab SDK):

- **PlayFab Authentication & Title Data:** Mengautentikasi identitas unik pemain (Custom ID / Login Anonymous) serta men-serialize status progres permainan ke dalam PlayFab Title Data.
- **Global Leaderboard Service:** Menjadwalkan dan mengagregasi skor pemulihan oksigen global (*globalO2Percentage*) serta waktu penyelesaian tahap (*stage completion time*) ke dalam papan peringkat global secara *real-time*.

BAB II: SETUP LINGKUNGAN PENGEMBANGAN DAN KOMPILASI BUILD

2.1 Persyaratan Perangkat Lunak (Prerequisites)

Sebelum membuka dan meng-compile source code proyek gim, lingkungan pengembangan berikut harus disiapkan:

1. **Unity Hub & Engine:** Unity 6 (6000.0.x) atau Unity 2022.3 LTS dengan modul:
 - Microsoft Visual Studio / JetBrains Rider Integration
 - Windows Build Support (IL2CPP / Mono)
 - WebGL Build Support
2. **Git Version Control System:** Terinstal pada OS untuk manajemen repositori.
3. **PlayFab SDK Package:** SDK PlayFab Unity v2.x terintegrasi di folder **Assets/PlayFabSDK**.

2.2 Panduan Setup dan Pengoperasian Project

1. Clone Repositori:

```
git clone https://github.com/username/life-on-land.git
cd "My project"
```

2. Buka di Unity Editor:

- Jalankan Unity Hub -> pilih **Add project from disk**.
- Arahkan ke folder root **My project**.
- Pilih Editor Version Unity 6 / 2022.3 LTS dan tunggu proses *asset importing* selesai.

3. Konfigurasi PlayFab Manager:

- Buka menu **Window** -> **PlayFab** -> **EdEx** (Editor Extensions).
- Masukkan **Title ID** PlayFab (misal: **1A2B3**).
- Pastikan aset **PlayFabSharedSettings** di **Assets/PlayFabSDK/Shared/Public/Resources/** telah terhubung dengan Title ID yang valid.

2.3 Prosedur Kompilasi (Build Pipeline)

1. Pilih Scene Permainan:

- Buka window **File** -> **Build Settings**.
- Tambahkan scene ke dalam daftar *Scenes In Build* sesuai urutan:

1. **Assets/Scenes/MainMenu.unity** (Index 0)
2. **Assets/Scenes/Stage1_RedRegion.unity** (Index 1)
3. **Assets/Scenes/Stage2_OrangeRegion.unity** (Index 2)
4. **Assets/Scenes/Stage3_PinkBloom.unity** (Index 3)

2. Kompilasi Standalone Windows (PC):

- Target Platform: **Windows**, Architecture: **x86_64**.
- Klik tombol **Build** -> Buat folder tujuan **Builds/Windows/** -> Eksekusi **LifeOnLand.exe**.

3. Kompilasi WebGL (Browser):

- Target Platform: **WebGL**.
- Color Space: **Gamma** (atau Linear jika URP WebGL diaktifkan).

- Klik tombol **Build** -> Pilih folder tujuan **Builds/WebGL/**.

BAB III: SPESIFIKASI SKRIP LOGIKA UTAMA (C# CLASS CONTRACTS)

3.1 Skrip Player.cs / PlayerController.cs

Mengontrol pergerakan fisik 2D, pemakaian stamina, buffer udara, dan interaksi perkakas hotbar.

```
public class Player : MonoBehaviour {
    [SerializeField] private string playerName = "Restorer";
    [SerializeField] private float baseMovementSpeed = 5.0f;
    [SerializeField] private float currentStamina = 100.0f;
    [SerializeField] private float localO2Buffer = 10.0f;
    [SerializeField] private int currentWaterInventory = 0;
    [SerializeField] private int activeHotbarSlot = 0; // Slot 0 s.d 5 (Tombol 1-6)

    public void ProcessMovementInput(float horizontal, float vertical);
    public void SelectHotbarSlot(int slotIndex);
    public void UseActiveTool(Vector2 targetGridCoordinates);
    public void ExecutePlantAction(TreeProfile profile, Vector2 targetGridCoordinates);
    public void ExtractWater(WaterSourceNode source);
    public void ConstructInfrastructure(BuildingBlueprint blueprint, Vector2
originCoordinates);
    private void EvaluateCalculatedDebuffs();
}
```

3.2 Skrip EnvironmentManager.cs

Mengelola siklus detak lingkungan (*environment tick*), difusi oksigen spasial, dan penguapan kelembapan tanah.

```
public class EnvironmentManager : MonoBehaviour {
    [SerializeField] private float globalO2Percentage = 15.0f;
    [SerializeField] private float tickInterval = 5.0f;
    [SerializeField] private bool isHeatwaveActive = false;

    private GridWorldMatrix environmentGrid;

    public void RecalculateAtmosphericComposition();
    public void ExecuteStateTick();
    public void DiffuseOxygen();
    public void DeployLocalizedDisasterEvent(int levelMilestone);
    public bool EvaluateVictoryState();
}
```

3.3 Skrip GridWorldMatrix.cs dan GridCell.cs

Mewakili dunia 2D berbasis sel matriks terstruktur untuk variabel tanah dan kontaminasi.

```
public class GridCell {
    public float moisture;           // 0.0f (kering) s.d 1.0f (basah)
    public int corruptionState;      // 0 = Normal, 1 = Burnt, 2 = Dug Burnt
    public float localO2;           // Kadar O2 pada sel ini
    public Tree plantedTree;        // Referensi komponen tanaman
}
```

```

}

public class GridWorldMatrix : MonoBehaviour {
    public int TilesPurifiedCount { get; private set; }

    public bool PurifyTileShovel(Vector2Int gridPosition);
    public bool PurifyTileWater(Vector2Int gridPosition);
    public GridCell GetCell(Vector2Int gridPosition);
}

```

3.4 Skrip Tree.cs dan GrowthState.cs

Mengontrol mesin status berhingga (*Finite State Machine*) pertumbuhan vegetasi.

```

public enum GrowthState { Seed, Sprout, Sapling, Young, MatureTree, Withered }

public class Tree : WorldObject {
    [SerializeField] private GrowthState currentFSMState = GrowthState.Seed;
    [SerializeField] private float localO2EmissionRate = 0.5f;
    [SerializeField] private int ticksSinceLastWatered = 0;
    [SerializeField] private int thresholdWaterRequirement = 3;

    public void ProgressGrowthCycle();
    public void InjectAtmosphericO2();
    public void TransitionToWitheredState();
    public void Revive();
}

```

BAB IV: SKEMA DATABASE DAN PERSISTENSI DATA

4.1 Skema Kontrak Data JSON (SaveData.json)

Seluruh state permainan yang perlu dipertahankan disimpan dalam format terstruktur berbasis JSON:

Field Key	Type Data	Deskripsi Fitur
playerId	string	Identifier unik pemain (PlayFab Login GUID / Local ID)
playerName	string	Nama akun pemain (default: "Restorer")
currentStamina	float	Sisa nilai stamina pemain (0.0 – 100.0)
localO2Buffer	float	Buffer cadangan oksigen lokal (0.0 – 10.0)
currentLevel	int	Tahap stage aktif (1 = Red, 2 = Orange, 3 = Pink Bloom)
inventory	Array<Object>	Daftar item inventoris { itemID , quantity }
gridCells	Array<Object>	Array koordinat sel dimodifikasi { x , y , moisture ,

		<code>corruptionState, growthState }</code>
<code>achievements</code>	Object	Flag pencapaian { <code>firstSteps: bool, waterBearer: bool, greenOasis: bool</code> }
<code>globalO2Percentage</code>	float	Nilai persentase oksigen global bumi terakhir tersimpan
<code>lastSavedAt</code>	string (ISO 8601)	Stempel waktu penyimpanan untuk validasi konflik cloud save

BAB V: PIPELINE ASSET DAN SISTEM DIALOG NOVEL VISUAL

5.1 Standar Pengolahan Grafik Pikel (Pixel Art Standards)

- **Pixels Per Unit (PPU):** Seluruh aset sprite dan tileset diset tepat ke **32 PPU**.
- **Filter Mode:** Wajib menggunakan **Point (no filter)** agar piksel tidak buram saat di-scale.
- **Compression: Uncompressed** (RGBA 32-bit).
- **Pivot Location:** Menggunakan **BottomCenter** pada prefab objek interaktif agar Y-sorting depth rendering bekerja sempurna terhadap posisi pemain.

5.2 Sistem Dialog Novel Visual (`DialogueManager.cs`)

- Memiliki panel UI overlay di bagian bawah layar dengan dukungan potret ekspresi karakter (Umbra, Maliz, Oryel, Pyper, Blaze).
- Berjalan otomatis saat pemain mendekati NPC atau memicu milestone tahap (*Stage Completion Gate*).